

Computer Organization and Architecture

Chapter 1 Introduction

Computer Architecture

- Attributes visible to the programmer
 - e.g. - instruction sets
 - I/O mechanisms
 - addressing techniques
 - number of bits used for data representation
- All Intel x86 or IBM/ 370 family share the same basic architecture

Computer Organization

- How features are implemented
 - e.g. - Memory technology
 - control signals
 - interfaces
- Organization differs between different versions

Structure (Organization)

- the way in which components relate to each other
 - Top Level (CPU, Memory, I/O, Systems interconnection)
 - CPU (ALU, Control Unit, Registers or high-speed memories, Internal CPU interconnection)
 - ALU (Registers and decoders, control memory)

Function (Architecture)

- the operation of individual components as part of the structure

All computer functions are:

- data processing
- data storage
- data movement
- control

Architecture VS Organization

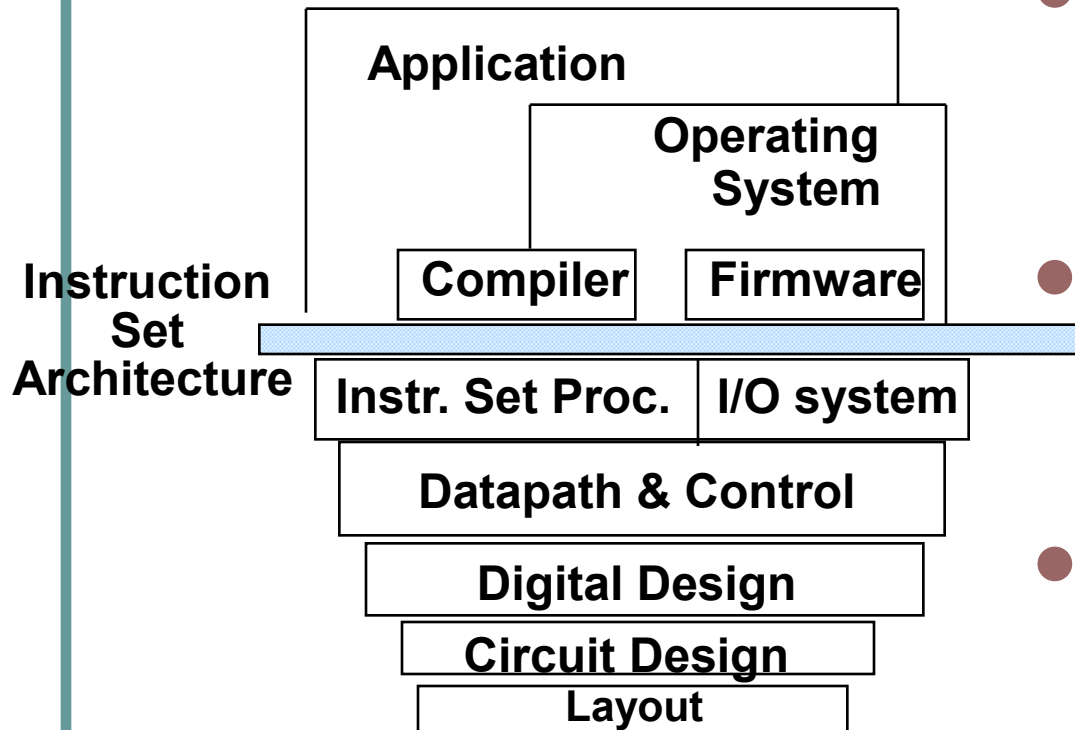
Architecture (what to do)

- Refers to the design of the internal workings of a computer system
- Concerned with optimizing the performance of a computer system and ensuring that it can execute instructions quickly and efficiently

Organization (how to do)

- Refers to the operational units (implementation) and their interconnections that implement the architecture specification
- Deals with how the components of a computer system are arranged and how they interact to perform the required operations.

What is “Computer Architecture”?



- Coordination of many *levels of abstraction*
- Under a rapidly changing set of forces
- Functional Design, Measurement, *and* Evaluation

WHAT TO DO

Question:

- Which of the following choices best describes the primary focus of computer organization compared to computer architecture?
 - a) Computer organization deals with high-level programmer-visible attributes like instruction sets, while architecture deals with physical components.
 - b) Computer organization defines the operational hardware units and how they realize architectural specifications, whereas architecture defines the functional behavior and logical design seen by the programmer.
 - c) Computer organization refers to software design choices, while architecture refers exclusively to peripheral hardware attachments.
 - d) There is no difference; both terms describe the physical manufacturing cost of a microprocessor chip.
- Answer: b

Machine Language

Machine Language  **Virtual Machine Language**

Machine language is the language understood by a computer. It is very difficult to understand, but it is the only thing that the computer can work with. All programs and programming languages eventually generate or run programs in machine language

- A machine language defined by a language out of an electronic circuit
 - raw, low-level binary code executed directly by a physical computer's CPU
- A machine with HPL like C/C++, Java, Modula 2
 - intermediate code meant for a software-based emulator or runtime, which then translates it for the physical hardware

Multilevel Machine Principles

- Layer/ Levels
- Translation – produce new program
- Interpretation – line by line
- Designing new computers or new levels must be familiar with levels other than the top ones

Multilevel Machine

- Level 6 and above - machines tailored to certain applications
- Level 5 – Problem Oriented Language level
- Level 4 – Assembly Language Level
- Level 3 – Operating System Machine Level
- Level 2 – Conventional Machine Level
- Level 1 – Microprogramming Level
- Level 0 – Digital Logic Level

Multilevel Machine

Level 6 and above



Translation

Level 5



Translation (compiler)

Level 4



Translation (assembler)

Level 3



Partial Interpretation (OS)

Level 2



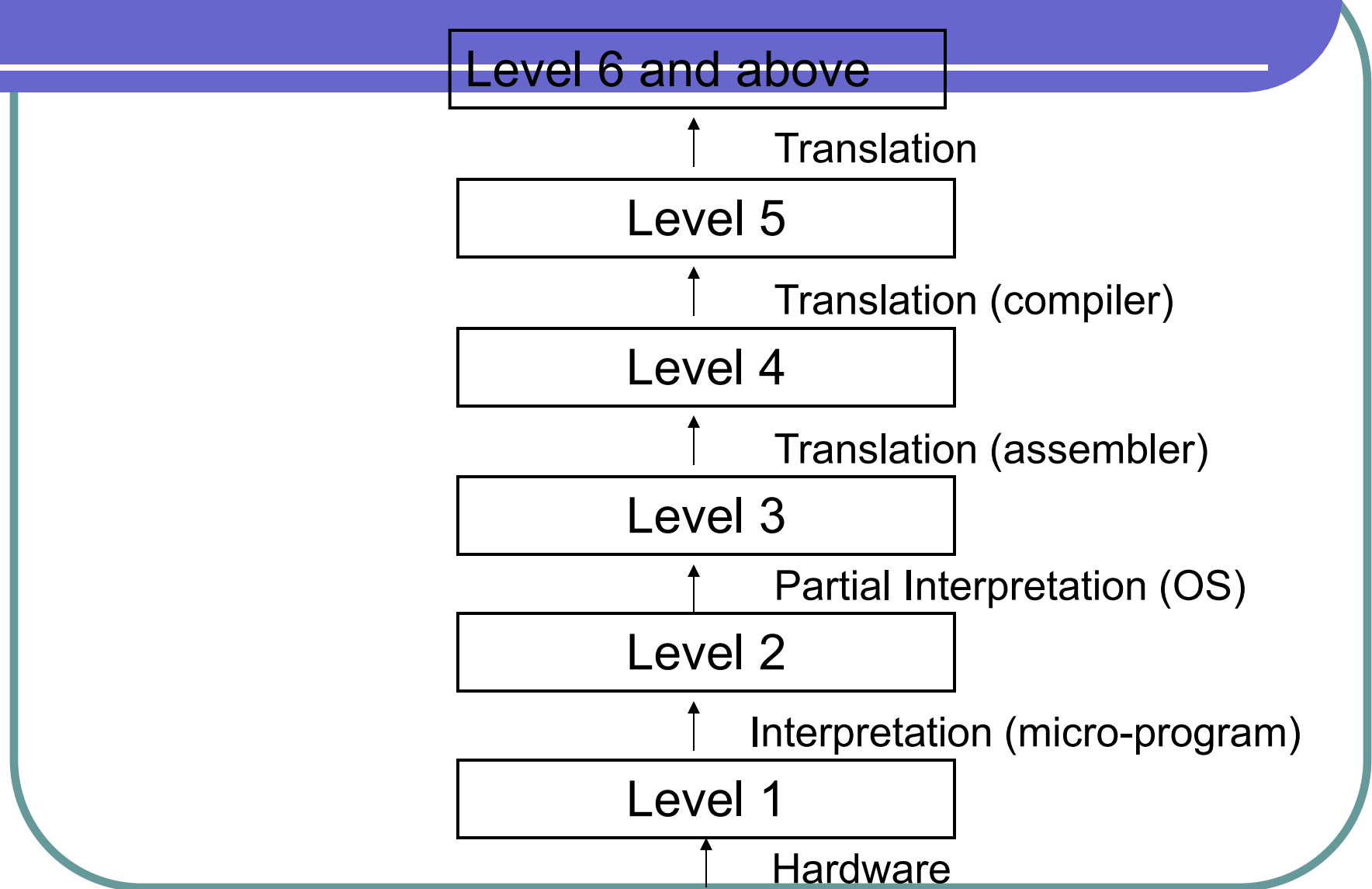
Interpretation (micro-program)

Level 1



Hardware

Level 0



Digital Logic Level (Level 0)

- machine's true hardware
- it's circuitry carries out the machine language programs of level 1
- Interesting objects are GATES

Microprogramming Level (Level 1)

- true machine level language
- A definite program (MICROPROGRAM) whose job is to interpret the instructions of level 2 (each micro-program implicitly defines a level 2 language)
- directly executed by the hardware

Conventional Machine Level (Level 2)

- MANUFACTURERS describing the machine's instruction set
 - instructions carried out interpretively by the micro-program
 - though some computers don't have a microprogramming level interpretation (micro-program)

Operating System Machine Level (Level 3)

- implemented by systems programmers
- hybrid level” – most of the instructions are also in the level 2
- additional features like memory organization, ability to run 2 or more programs in parallel, etc, are included
- some are interpreted by the OS and some instructions are interpreted by the microprogram

Assembly Language Level (Level 4)

- intended for the applications programmers with problems to solve
- change – the method by which the higher levels are supported (supported by translation)
- Languages contain words and abbreviations meaningful to people

Problem Oriented Language Level (Level 5)

- Consists of languages designed to be used by applications programmers with problems to solve (HLL)
- Translation (compiler) – translates level 5 to level 3 or 4 (although, occasionally they are interpreted)

Level 6 and Above

- consist of collections of programs designed to create machines specifically tailored to certain applications (e.g. administration, education, computer design, etc.)
- these levels are in area of research and development

Different classification of today's Computer

- **Microcomputers**

- smallest, least costly, and most popular computers on the market. The availability of low-cost, easy-to-use programs plays a major role in consumer acceptance of microcomputer brands.

Different classification of today's Computer

- Minicomputers

- performs specialized tasks such as handling data communications. Smaller, cheaper, easier to maintain and install than mainframes. (Quickly disappearing because micros can now perform what minis can do).

Different classification of today's Computer

● Mainframes

- are large, fast systems capable of supporting several hundred input and output devices. Frequently used as repositories of huge amount of data.
- Staff operates and programs the mainframe computer.

Different classification of today's Computer

- Supercomputers

- are the fastest most expensive computers.
- Can run different calculations simultaneously, thereby processing in a minute what would take PCs several weeks or months.
- Most of them are used for scientific research, weather forecasting, pollution/environment condition analysis, etc.

References

- William Stallings, Computer Architecture and Organization. 11th Edition, 2019
- Structured Computer Organization by Andrew Tannenbaum